Dissecting 5G in the Wild: Performance, Coverage, and Support for Next-Gen Applications

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Abstract

5G, officially rolled out in 2019, has been rapidly deployed by major mobile network operators across the globe. Key advancements in 5G such as higher-order modulation, massive MIMO, wider channels, and higher operating frequencies, have set high expectations over the predecessor technology LTE, especially for enabling high-bandwidth and latency-sensitive applications such as Mixed Reality (XR), Connected Autonomous Vehicles (CAVs), and 360° video streaming — often referred to as 5G "killer" apps. While 5G's technical promises have fueled widespread excitement, understanding its true impact requires a closer look at the real-world performance delivered to end users.

This abstract highlights our ongoing efforts over the past four years to systematically evaluate the performance and capabilities, and uncover different operator policies of operational 5G networks through extensive out-in-the-wild measurement campaigns. We examine network coverage, throughput, latency, and emerging features, as well as 5G's interaction with the "killer" apps. Our findings reveal both the strengths and limitations of current 5G deployments. Building on these insights, we also propose and implement solutions aimed at addressing key challenges, with an eye toward shaping the design of future 5G and Beyond-5G (B5G) systems.

CCS Concepts

• **Networks** → **Mobile networks**; **Network measurement**; **Network performance evaluation**.

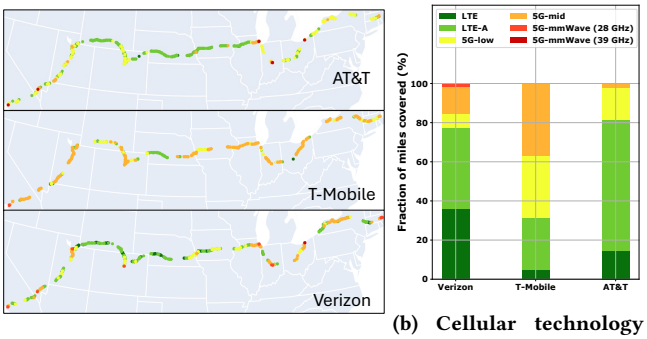
Keywords

5G, Measurement, Driving, Coverage, Network Performance, Application Performance, Dataset

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1 Measurement-driven Evaluation of 5G Networks in the Wild

Performance of cellular network on the wheels. In 2022, we collected a first-of-its-kind multi-carrier, multi-technology, multi-band, cross-layer cellular network dataset through a cross-continental US driving trip (5700km+) [2]. Fig. 1a shows the measurement route and the cellular coverage map for all three major US operators: Verizon, AT&T, and T-Mobile. Fig. 1b shows the technology coverage for all three operators as a percentage of miles driven during the entire trip. Despite the 5G rollout in 2019 and aggressive deployment efforts over three years (2019-2022), *5G coverage was in general disappointingly low and highly fragmented across the country*. T-Mobile had the highest % of 5G deployment (around 70%); in contrast, with Verizon and AT&T, we were able to connect to 5G less than 25% of the total distance travelled. In addition, the percentage of high-speed 5G (midband and mmWave) was even lower ranging from 38% (T-Mobile) to as low as 3% (AT&T).



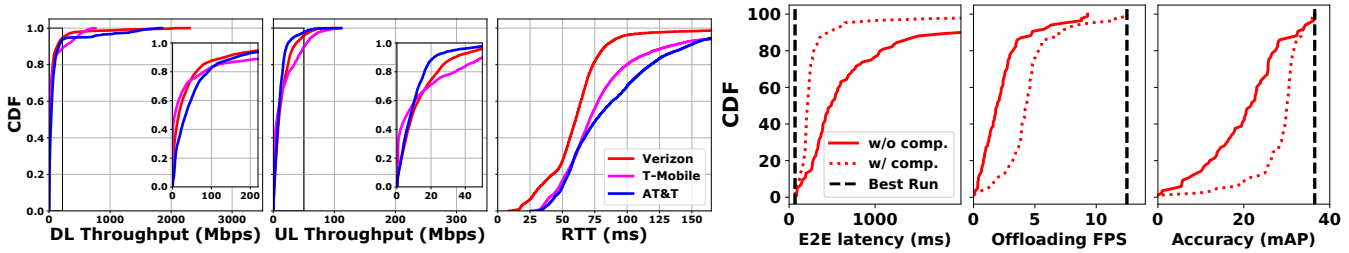
(a) Coverage map. **(b) Cellular technology breakdown per operator.**

Figure 1: Cellular network coverage statistics for 3 major US operators from our 2022 cross-country measurement campaign.

Our results in Fig. 2a showed degraded performance in terms of throughput and latency for all 3 operators while the users were on the move. All three operators demonstrated impressive peak speeds — over 2 Gbps downlink and 100+ Mbps uplink — thanks to mmWave deployments and wider channels. However, median and 75th percentile downlink performance dropped to just 6–34 Mbps and 47–74 Mbps, respectively. Uplink performance ranged from 6–9 Mbps (median) to 14–24 Mbps (75th percentile). We also observed inflated RTT values, with occasional spikes as high as 2–3 seconds across all three operators.

Such performance degradations have a direct impact on application-level experience. We built a canonical edge-assisted single user AR app that offloads compressed and uncompressed camera frames (30 FPS) over the cellular network to a cloud/edge server to perform a DNN-based object detection. We evaluated AR performance using three metrics: (a) the average E2E offloading latency, (b) number of offloaded frames per second, and (c) the accuracy of the object detection in terms of mean average precision (mAP). Fig. 2b, shows the performance of the AR app while driving. *We observed a significant performance drop during mobility — with higher E2E latency compared to the best-case static scenario (marked by a dashed line), and correspondingly lower offloading frame rate and detection accuracy.*

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(a) Throughput and RTT per operator while driving.

(b) Performance of AR app.

Figure 2: Network and application performance from our 2022 cross-country measurement campaign.

In 2024, we conducted a follow-up cross-continental measurement campaign to track how 5G coverage and performance have evolved. Initial results show substantial improvements in coverage for all three operators. T-Mobile's 5G coverage grew from 70% to 94%, while Verizon and AT&T also improved, but their 5G coverage still hovers around 50%. We are currently continuing our head-to-head analysis of the 2022 and 2024 datasets to compare coverage, performance, and network behavior over time. In parallel, given the rising interest in Non-Terrestrial Networks (NTNs) as a viable complement or substitute for cellular connectivity, we have also begun evaluating Starlink (LEO satellite) performance in 2024 to assess how it stacks up against terrestrial networks.

Resource allocation policies in operational 5G mmWave networks. We took a first look at resource allocation policies in operational 5G mmWave networks [1] by performing extensive experiments across 4 major US cities with 2 major operators. We found that resource sharing in 5G mmWave networks is strictly governed by the operator and multiple flows do not directly compete against each other in a shared queue. Mobile operators neither employ the well-known proportional fairness policies (which are considered the de facto standard for opportunistic schedulers in cellular networks) nor do they aim at maximizing the total throughput. Instead, they leverage threshold-based resource allocation policies based on user traffic demands and often over-allocate resources to new flows or reserve some capacity for future use.

Ongoing and future 5G measurement works. We are currently demystifying operator-specific handover (HO) configurations in the 5G era. Some of our preliminary findings are as follows: (a) The 5G HO landscape is very different from the LTE landscape. (b) Despite the complexity of the 5G ecosystem, the richness of HO configuration values appears reduced compared to LTE. (c) Our preliminary analysis shows that the selection of optimal configurations remains very challenging for operators, as they try to strike a balance between better post-HO performance and timing (which can deteriorate pre-HO performance), while also mitigating the "ping pong" effect.

In parallel, we are conducting the first large-scale study of 5G Standalone (SA) deployment in the US, analyzing its evolution while comparing it head-to-head with 5G Non-Standalone (NSA). Preliminary results suggests a shift in 5G deployment priorities from 2023 to 2024 in favor of 5G SA as compared to 5G NSA, with 5G SA deployments extending across diverse geographical areas and timezones. We noticed that advanced features, such as higher carrier aggregation with diverse band combinations and new frequency bands with different duplexing capabilities, allow SA to achieve better performance compared to NSA, whose peak performance has plateaued, reflecting limited upgrades as operators transition to SA infrastructure.

2 Solutions for Next-Gen Applications

The inflation in the E2E latency in the uplink-oriented AR application, as shown in §1, is a combination of multiple factors: frame transmission time, network RTT, and DNN inference time. While the network RTT and the DNN inference time are influenced by the server location and the compute capabilities of the server, the frame transmission time is directly dependent on the underlying wireless network. We found that the high frame transmission time results from the inherent asymmetry between downlink and uplink performance in current cellular deployments. This asymmetry is mainly a consequence of static, downlink-heavy PHY layer TDD configurations, a legacy design choice better suited for traditional downlink-centric applications, but poorly aligned with the needs of emerging uplink-intensive applications. This highlights the need for traffic-aware scheduling policies that reflect modern application demands.

To address this problem, we proposed a dynamic TDD configuration approach [3], where the user equipments (UEs) perform channel estimation for a given time window and report it to the network along with their traffic demands. The network, based on the channel metrics and traffic demands of the UEs, dynamically allocates resources for each UE so that the UEs are able to finish the tasks within their deadlines. We implemented this system on a 3GPP compliant testbed with two UEs having heterogeneous traffic demands and traffic types. Our results show that the dynamic TDD configuration system satisfies the overall traffic demands of the two users more often than any static TDD configuration. At the same time, it is left with the lowest percentage of untransmitted bytes at the end of a given latency-deadline cycle. This work serves as a proof-of-concept demonstrating the feasibility and benefits of traffic- and channel-aware dynamic TDD allocation in 5G networks. We are currently working to scale this solution to multi-user scenarios, while accounting for inter-cell interference from neighboring base stations.

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